

Pipeline Hazards & Resolution

Q. Define Pipeline Hazards. Classify and explain the three main types of pipeline hazards (Structural, Data, and Control) and briefly mention methods to handle them.

****Definition to Remember****

A ****Pipe**
pipeline

1. Structural Hazards

Occurs when two instructions in the pipeline need the same hardware resource at the exact same time (Resource Conflict).

* ****Ex**
memory

2. Data Hazards

Occurs due to Data Dependency, meaning an instruction needs the result of a previous instruction that hasn't finished executing yet.

* ****Ex**

3. Control (Branch) Hazards

Occurs due to Branch instructions (Jump, If-Else). The pipeline fetches instructions sequentially. When a branch is encountered, the target address isn't known until the execute stage. Meanwhile, wrong instructions may have been fetched.

* ****Ha**

****Must-Write Points (for 10 marks)****

1. ****Pipe**

*****Quick Recall*****

*`Pipeline`
Forward*